Blending Two Coffees

A worked example of a linear system

1 The Problem

A café mixes Arabica beans, which cost 400 baht per kilogram, with Robusta beans, which cost 250 baht per kilogram. It needs 10 kg of a blend that costs 3,400 baht in total. Let x and y be the masses of Arabica and Robusta in kilograms. Then

$$x + y = 10, \tag{1}$$

$$400x + 250y = 3400. \tag{2}$$

Equation (1) fixes the total mass and equation (2) fixes the total cost.

2 Solution by Substitution

From (1) we get $y = 10 - x$. Substituting into (2) gives

$$\begin{aligned} 400x + 250(10 - x) &= 3400 \\ 400x + 2500 - 250x &= 3400 \\ 150x &= 900 \\ x &= 6, \end{aligned}$$

and therefore $y = 10 - 6 = 4$. The blend uses 6 kg of Arabica and 4 kg of Robusta.

3 Solution with Matrices

The same system can be written as $A\mathbf{v} = \mathbf{b}$, that is

$$\begin{bmatrix} 1 & 1 \\ 400 & 250 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 3400 \end{bmatrix}. \tag{3}$$

The determinant of the coefficient matrix is not zero,

$$\det A = \begin{vmatrix} 1 & 1 \\ 400 & 250 \end{vmatrix} = 250 - 400 = -150 \neq 0,$$

so (3) has exactly one solution. By Cramer's rule,

$$\begin{aligned} x &= \frac{1}{\det A} \begin{vmatrix} 10 & 1 \\ 3400 & 250 \end{vmatrix} = \frac{2500 - 3400}{-150} = 6, \\ y &= \frac{1}{\det A} \begin{vmatrix} 1 & 10 \\ 400 & 3400 \end{vmatrix} = \frac{3400 - 4000}{-150} = 4, \end{aligned} \tag{4}$$

which agrees with Section 2: $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$.

4 Delivery Charge

The café delivers the blend for a charge that depends on the mass m in kilograms:

$$C(m) = \begin{cases} 80 & \text{if } 0 < m < 1, \\ 50 & \text{if } 1 \le m < 5, \\ 0 & \text{if } m \ge 5. \end{cases} \tag{5}$$

An order of 10 kg is therefore delivered free of charge.